

BeadMuse AI: Enhancing Inclusive and Independent Crafting Through Adaptive Pixel Art Templates

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ABSTRACT

The pursuit of empowerment and democratization in the maker movement necessitates a stronger focus on inclusivity towards individuals with disabilities. Despite advancements in maker technologies and environments, a gap persists in exploring the accessibility of creative craft practices. This collaborative study was carried out with a company dedicated to inclusive employment focused on understanding the challenges of beginners with intellectual disabilities approaching the use of iron beads as a creative crafting medium. Leveraging insights from a workshop involving five company employees, we developed the BeadMuse AI—a web application system that enriches the crafting experience by tuning into users' interests to produce custom templates that align with individual strengths and preferences. Preliminary evaluation shows the potential for BeadMuse AI to support independent crafting among disabled employees, with potential for broader applications.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in accessibility*.

KEYWORDS

Accessibility, Intellectual disability, Creative crafting, Inclusive employment

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1 INTRODUCTION

In recent years, there has been a growing focus within HCI on exploring the challenges of inaccessibility faced by people with disabilities in the maker movement [1, 17]. While significant efforts have been made to enhance the accessibility of high-tech electronics and digital tools, there is also a growing recognition among scholars that traditional crafting practices have received limited attention in terms of inclusivity for people with disabilities [3, 9]. Previous research has pointed out how traditional crafting practices such as woodworking, weaving [3], and modern activities like soldering [19], often demand specific spaces and equipment as well as trained instructors. This paper examines a unique form of creative art crafting, known as iron beads crafting, which combines traditional and contemporary elements of making and offers a more leisurely and accessible approach for people with intellectual disabilities (ID) to engage in crafting.

Iron Beads. Iron beads crafting stands as a versatile and accessible medium within the realm of creative practices. This craft involves arranging and fusing small plastic beads using heat to create intricate designs and patterns. Its simplicity and extensibility make it suitable for individuals of varying skill levels. The materials used in iron-beads crafting include beads of different sizes, pegboards, and an iron for fusing the beads. See Figure 1 for detailed steps.



Figure 1: Three steps of Iron Beads Crafting

Collaboration with Company Yakuju Will. Our collaboration takes root in Yakuju Will¹, a distinct subsidiary of a Japanese enterprise we shall abbreviate as "Company Y". Dedicated to fostering an inclusive workspace, Company Y employs 32 individuals (as of October

¹Yakuju Will website: <https://yakuju-will.com/>

2023), of which the majority, 24 employees, have varying degrees of ID. Despite their challenges, these employees proficiently handle their daily tasks which encompass cleaning, folding laundry, and managing cloth logistics. Their cloth-related duties have honed their expertise in using equipment such as irons. Seeing this, we identified an opportunity in iron beads crafting, enabling them to utilize their familiarity with these tools to create personalized artistic expressions. Noting that the individuals with ID we collaborated with are employed, distinguishing them from the conventional perception of people with ID. This underscores their capabilities and sets a unique context for our study.

In this article, we first present and analyze the process and findings stemming from a user-centered iron beads crafting workshop involving the main participation of 5 individuals with ID from Company Y, supported by facilitators who provided valuable support throughout the creative journey.

Employing the methodology of semi-structured group discussions, observations, and post-surveys, we uncovered the depth of the participants' engagement in self-expression and creativity during the bead crafting process. Moreover, we noted the impact of varying skill levels on the crafting process and final results, along with the participants' reliance on external references for image-to-template conversion.

Subsequently, building upon the insights garnered from this workshop, we introduce BeadMuse AI—an innovative AI-driven web application system crafted to empower individuals with ID to engage in independent crafting. This platform offers an interactive space where users with ID can seamlessly translate their personal preferences and creativity into intricate bead crafting templates. Importantly, BeadMuse AI is designed to operate effortlessly, either within the company's premises or during leisure hours, fostering free opportunities for imaginative crafting, thus negating the necessity for structured workshops.

Finally, we presented BeadMuse AI to the company's management team and facilitated their hands-on interaction with the system. Through semi-structured interviews, we obtained an assessment from the management regarding the potential application of BeadMuse AI in real workplace settings. They provided two suggestions for design improvements, which will further enhance the system. Moreover, during these interviews, they elaborated on various potential usage scenarios, reaffirming the extensive potential of BeadMuse AI.

2 RELATED WORK

2.1 Accessibility in Crafting and Making in HCI

Within the realm of making and fabrication, researchers have diligently pursued the creation of accessible tools and have recognized the significance of configuring materials within maker spaces to foster a sense of empowerment in makers [15]. A few illustrative examples underscore this trend. Siu et al. developed a system focused on assisting individuals with visual impairments. Their approach centered on enabling users to manipulate and adjust 3D models through tactile interactions via a 2.5D shape display [22]. Similarly, Denner et al. evaluated the potential of the video game, Minecraft, to function as an assistive tool for young individuals with autism.

Their study highlighted the possibility of adapting existing technologies to meet distinct needs within the disability community [16]. Extending this line of work, Katya et al. incorporated audio-enhanced technologies to facilitate weaving tasks for those with visual impairments [3]. Despite these advances, gaps remain in the literature. Specifically, there's a discernible need for further research into traditional crafting practices, especially in relation to their accessibility dimensions [3].

2.2 Address Independence in Disabilities

Independence, in the context of individuals with disabilities, pertains to the ability to exercise control over personal decisions and daily routines [5]. This idea intersects with the notion of self-determination, which is fundamentally about possessing agency in one's life, taking proactive actions to sustain or enhance its quality, and encompasses decision-making, goal setting, and attainment, problem-solving, as well as self-awareness, advocacy, and efficacy [23]. Rather than being limited to the confines of their environment, which might exacerbate feelings of anxiety and reduce self-esteem [2, 14]. This has led to the development and research into numerous assistive technologies specifically designed to reinforce such autonomy among individuals with disabilities. A lot of studies have extensively explored the utilization of assistive technologies to foster a sense of independence among individuals with disabilities. For instance, innovations in 3D printing for educational purposes and mobile platforms have been identified as promising avenues in this endeavor [6, 8]. To evaluate the efficacy of these assistive technologies in fostering independence among individuals with disabilities, the self-determination theory provides a robust framework [10]. By examining autonomy, competence, and relatedness, researchers can ascertain the degree to which these tools meet the desired outcomes for their target audience.

2.3 Benefits of Crafting for People with Disabilities

Crafting activities have been extensively studied and are recognized for their multiple benefits. They have been shown to promote well-being and health [7] and play a significant role in facilitating self-development [12]. Furthermore, engaging in crafting can bolster an individual's sense of identity, enhance self-awareness, and foster a feeling of belonging within a community [20]. Beyond these psychological and social benefits, crafting activities can also serve as a therapeutic medium to alleviate stress and enrich leisure time [18]. In Risatti's analysis, crafting produces tangible objects that are both functional and aesthetic, thereby serving as mediums of self-expression [21]. Positioning crafting within a more inclusive context, Flood et al. assert that for older adults, crafting extends beyond a mere hobby. It becomes a creative endeavor that intertwines cognitive stimulation with emotional fulfillment. In this scenario, crafts are not merely for relaxation but also play a pivotal role in maintaining cognitive and emotional health [11]. When viewed in the context of individuals with intellectual disabilities, crafting offers an avenue to showcase abilities, thereby contributing to a broader societal impact. The act of crafting and subsequently

'gifting' these creations becomes a conduit for transmitting emotions and creativity into nondisabled spaces, fostering avenues for connection and recognition [13].

3 WORKSHOP STUDY: METHODOLOGY

Our exploration into enhancing the accessibility and independence of iron beads crafting practices specifically for individuals with ID from Company Y commenced with a designed user-centered iron beads workshop. This approach allowed us to glean insights directly from the participants' experiences, shaping our following design to harmonize with their crafting needs and preferences. The workshop session was held in July 2023 and lasted for 5 hours. See Figure 2.



Figure 2: (A): Pixel transformation by facilitator (B&C): Participants place beads on pegboard based on the template (D): Participants take photos with their crafted items

3.1 Purpose Setting

Participants attended the workshop accompanied by the entire Company Y management team. The objectives of the workshop encompass three primary dimensions, aimed at comprehensively understanding the needs, capabilities, and challenges of individuals with intellectual disabilities in the creative process, with the overarching goal of enabling long-term independent crafting both within the realm of the company's supported leisure activities and beyond, even in the absence of trained facilitators.

Observing Self-Expression and Artistic Creation: Participants were asked to bring along a photo of an item or scene that held personal significance or evoked a strong emotional connection. Through this exercise, we aimed to understand not just their ability to replicate the imagery in bead form, but also to observe the emergence of their artistic creativity in the crafting process.

Identifying Barriers to Independent Crafting: We offered manual assistance and facilitated the whole process to identify areas where collaborative support was necessary, as well as pinpoint barriers to accessible and independent crafting. Finally, we sought to identify specific elements and barriers that hindered independent crafting.

Enabling Long-Term Independent Crafting: By leveraging the possible insights gained during the workshop, our ultimate aspiration

was to discover methods that would enable individuals to engage in crafting independently over the long term, without continuous facilitator presence.

3.2 Participants

P1 to P5: These individuals are employees from Company Y with varying levels of ID. They were the primary participants in the iron beads crafting workshop, engaging actively in the crafting process. Their crafting experience differed, which provided a wide spectrum of perspectives and challenges.

P6 to P9: Company Y management team played a dual role. They acted as supporters, assisting the primary participants (P1 to P5) throughout the crafting process. Additionally, they observed the workshop, which enabled them to provide insights from a managerial and operational perspective, specifically concerning the feasibility of replicating similar activities within the context of company-supported leisure activities. Their experience with iron bead crafting also varied, enriching the depth of observations and feedback obtained.

Table 1: Participants' demographic information

Participant	Gender	Disabilities and differences	Iron Beads Crafting Experience
P1	Female	Intellectual disability (mild)	created before
P2	Female	Intellectual disability (mild)	created before
P3	Male	Intellectual disability (mild)& ADD	know about it but never created before
P4	Male	Intellectual disability (mild)	know about it but never created before
P5	Female	Intellectual disability (moderate)	know about it but never created before
P6	Male	CEO of Company Y	know about it but never created before
P7	Male	Manager of Company Y	know about it but never created before
P8	Female	Manager of Company Y	created before
P9	Female	Manager of Company Y	created before

3.3 Facilitators

There were a total of five trained facilitators from our team for this workshop, each responsible for guiding and assisting participants to ensure the smooth progression of the activities and the achievement of the overall objectives. The facilitators' roles encompassed three essential responsibilities: (1) Image Conversion: Transforming participants' photos into pixel templates using iPads for iron bead crafting. (2) Bead Arrangement and Ironing Assistance: Guiding participants in placing iron beads on pegboards for accurate designs. Helping in the final step to ensure beads fused together properly. (3) Documentation: Capturing the crafting process and outcomes through photography and videography for future analysis. See Figure 3 for details.

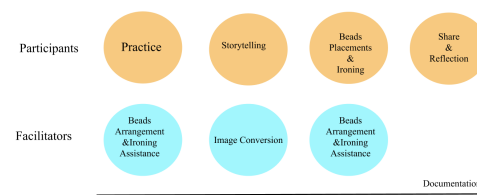


Figure 3: Process of workshop of both participants and facilitators

3.4 Procedure

The workshop commenced with participants practicing in groups, engaging in simple bead crafting exercises to familiarize themselves with the entire process. As they worked on these initial creations, they had the opportunity to become acquainted with the crafting materials and methods. After practice, participants were encouraged to storytelling the select images that resonated with them personally, meanwhile, as a pivotal step, facilitators utilized iPads to seamlessly transform participants' selected images into pixel templates. With templates ready, participants embarked on the crafting phase. Upon completing their bead arrangements, participants progressed to the ironing phase, fusing the beads together to form a cohesive piece. Finally, armed with toolkits and guidance, participants enriched the crafting experience by transforming their beadwork into functional accessories, such as keychains and fridge magnets. See Figure 4 for details.

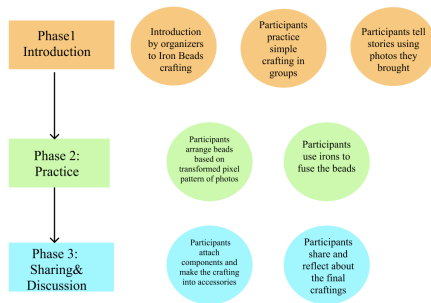


Figure 4: Workshop Procedure

Throughout the workshop, we captured the sessions on video for observational analysis. As it concluded, we initiated a semi-structured group discussion, distributed questionnaires to the nine participants, and conducted an in-depth interview with a management representative (p6). Our thematic analysis [4] revealed common trends in opportunities and challenges which we present in the following section.

4 WORKSHOP STUDY: FINDINGS

Opportunities emerged in the form of employees' self-expression through personalized crafting and the stimulation of creativity. Challenges, on the other hand, included varied obstacles faced by participants with different crafting experiences and a pronounced desire for independent crafting juxtaposed with a dependency on references.

Self-Expression through personalized items. "Trimming business cards is part of my job and I love my job, so I wanted to create something to represent it" P1 shared during the workshop. P2, who crafted a Pokémon pattern, also conveyed, "I really like Pokémon; I play its game in my free time." Our observations revealed that participants not only actively engaged in the crafting process but also utilized it as a means to convey their personal stories and preferences. Each participant enthusiastically shared their unique

narratives associated with their chosen crafts. This trend was further exemplified when P4 reflected in the post-survey that they "learned more about their peers."

Creativity Stimulation. Our observations and group discussions highlighted participants' proactive creativity. While facilitators provided templates, participants exhibited alterations to designs and colors. P3 shared, "I prefer pink, so I changed the apron's color from green to pink". P5 creatively suggested, "I think the hedgehog doesn't need a tail, it will look more harmonious". These observations underline the workshop's role in stimulating participants' creative thinking and maintaining their focus on the crafting process.

Varied Reception and Skill Levels. Our analysis uncovered diverse skill levels and varying responses to challenges among participants. Experienced participants (P1 and P2) sought to challenge themselves by opting for more intricate designs using smaller beads, resulting in the creation of multiple intricate bead crafts. In contrast, those with slower crafting speeds (P3 and P5) approached the process more cautiously. These observations were corroborated by post-survey responses, where P1 expressed a desire for more intricate challenges, while both P3 and P4 noted the challenging aspects of the workshop. This variability in reception and skill levels highlights the individualized approach required to engage participants effectively.

Dependency on Reference of Image-to-Template Conversion. A significant theme emerged regarding participants' reliance on image-to-template conversion. "Although you can find many existing patterns online, sometimes when you want to create something unique, you worry about not doing it well". P1 revealed, "I get confused and hesitate, not knowing which image to choose". This dependency on image-to-template conversion and unsure of chosen showcases the participants' need for accessible tools to facilitate their creative endeavors. These findings strongly reinforce our belief in the profound affinity individuals have for beads crafting and the significant assistance it offers them. Moreover, these insights underscore the need for our future design endeavors to be attuned to varying levels of abilities and the facilitation of their creative process through external references. The intimate connection between their fondness for beads crafting and its empowering effects guides our mission to create a platform, like BeadMuse AI, that enables them to engage in crafting anytime, anywhere, and with greater ease.

5 DESIGN OF BEADMUSE AI PROTOTYPES

5.1 Defining Functional Requirements

Based on the workshop's process documentation and data analysis, several aspects need to be considered for iron beads crafting for individuals with ID: how to generate personalized pixel template references, grading the level of crafting difficulty, designing UI interactions, and assisting in improving aesthetics.

Skill-Level Adaptation. Based on observations and analysis from the workshop, it's evident that participants faced persistent barriers to varying abilities when it comes to iron beads crafting. Some participants tend to choose simpler and larger beads, resulting in a



Figure 5: BeadMuse AI Web Interface

slower pace of crafting work. However, there are also a few participants who not only opt for smaller beads but also select images with higher levels of difficulty. Surprisingly, these participants completed the crafting work at a rapid pace. Furthermore, as participants gain more experience in iron beads crafting, their skills also improve. Therefore, establishing different skill levels for iron beads crafting is crucial. The interfaces allow for participants' skill levels to be categorized as beginner, intermediate, and advanced. Depending on the level of difficulty, BeadMuse AI will provide pixel images of varying complexities.

Scale Customization. The pixel image size generated from the original photo needs to be set based on participants' preferences, as different participants desire varying sizes for their handicraft creations. Simultaneously, consideration should also be given to the size of the beads chosen by the participants, ensuring that the converted pixel image fits within the dimensions of the bead board.

User-Friendly Interface. For the design of interactive interfaces aimed at patients with cognitive impairments, we strive to simplify the interaction steps as much as possible. In terms of language, we use concrete words instead of abstract adjectives. For example, we use 'new to this' instead of 'beginner' in an introductory manner, making it easier for patients to understand the interactive interface.

5.2 Addressing Accessibility Issues

This design, while referencing the Web Content Accessibility Guidelines (WCAG) and incorporating the manual processes from workshops, has created the web version of BeadMuse AI for easy user access. In the design of the interaction interface, BeadMuse AI employs clear and high-contrast colors. Enlarged and prominent buttons will be used for user clicks, ensuring a consistent visual style for each step to facilitate user comprehension. (see Figure 5 for whole prototype design) In the context of a workshop involving individuals with ID in the process of manual craft design, it was observed that during the initial stages, they faced challenges in determining what type of reference object to choose for creating pixel art. However, when facilitators provided reference images, participants reviewed pictures from their own mobile photo albums and discussed with people around them, they were able to draw inspiration and subsequently make decisions regarding their design objectives. Therefore, this study utilized the Wix² web development

platform and integrated OpenAI³ artificial intelligence image generator API along with a text-image-based GPT-3 language model. This combination was used to generate corresponding images based on the photos uploaded by ID and the accompanying descriptive text. The process identifies key objects present within the photos and generates multiple pixelated reference images of these objects that conform to the mosaic board dimensions. Subsequently, participants are invited to select their preferred images, thus aiding and inspiring them in expeditiously choosing their manual craft design objectives.

5.3 Technical Implementation

BeadMuse AI requires users to select their preferred difficulty level and the size of beads for beadwork through buttons. It records the user's button selections and translates them into corresponding keywords. Finally, it utilizes OpenAI's image generation API, offering users three options: (1) Generate AI images based on text input. (2) Generate AI images based on uploaded images. (3) Combine text and uploaded images to generate AI images. If users choose large-sized beads, the generated images will be divided into 29x29 square segments. Conversely, if they opt for small-sized beads, the generated images will be divided into 50x50 square segments. We refined the mobile interface (see Figure 6), observing that several participants autonomously turned to their mobiles for another round of pixel template searches. These AI-generated images for generation interface and examples serve as design references and creative inspiration for individuals with ID during the beads work process. If patients are dissatisfied with the generated images or seek further inspiration, they can choose to go back and regenerate the images.



Figure 6: Workflow showcase in a mobile device and take the prompt "a dog with tools" as an example in Japanese

²<https://www.wix.com/>

³<https://platform.openai.com/>



Figure 7: Left: Interviewee interacting with BeadMuse AI. Right: Screenshot of interface in Japanese.

6 DESIGN EXPLORATION: METHODOLOGY

We first introduced the BeadMuse AI system to the management of YakujuWill through a comprehensive showcase, followed by a semi-structured interview to assess expert perspective on the system's feasibility. Refer to Figure 7.

6.1 Interviewee

We engaged in discussions with the CEO of Company Y (p6), a central figure in the company's operations and closely connected to employees with ID. Beyond their administrative roles, the CEO had also been an active participant in our previous iron bead crafting workshop and he is a key promoter of the integration of creative activities as a way to promote a sense of engagement and belonging among company employees.

6.2 Procedure

The interview with the CEO of Company Y was conducted in person in September. Spanning a duration of one and a half hours, the session was video recorded. Alongside the in-person interactions, we also demonstrated the BeadMuse AI prototype on a computer, capturing the screen throughout to ensure all interactions with the system were documented.

After showcasing the BeadMuse AI prototype and allowing the CEO to interact with it firsthand, we initiated the semi-structured interview. The primary aim of these discussions was to gather the CEO's feedback on the prototype and understand their perception of its potential fit within the company. The questions included:

- What are your impressions of the BeadMuse AI prototype?
- How do you foresee the employees' reception of this prototype?
- In your view, what potential impacts might this prototype have on the employees' creativity, both positive and negative?
- Are there any elements of the prototype that you believe need modifications or enhancements?

Our subsequent analysis using thematic analysis[4] centered on evaluating the prototype's functionality, design, and its potential to support employees in their long-term iron beads crafting endeavors.

7 DESIGN EXPLORATION: FINDINGS

7.1 User Feedback and Recommendations

Website Design Feedback. The interviewee's feedback on the BeadMuse AI's design was overwhelmingly positive. He expressed

appreciation for the "clean", "intuitive" layout, suggesting it fostered a "conductive environment" for creativity. However, he also highlighted a few enhancement opportunities. A primary suggestion was to streamline user navigation through the inclusion of a progress bar. As he elaborated, *"The process entails merely three steps. While it appears straightforward, there's potential for our employees to feel disoriented. A progress bar specifying '2 more steps to go' could clarify the journey."* Furthermore, to ease beginners into the system, he proposed integrating visual cues. Specifically, showcasing images of crafted products across varying skill tiers would not only aid novices in their choices but also mitigate any hesitation stemming from unfamiliarity with the crafting genre. He remarked, *"For those versed at intermediate or advanced levels, navigation might be effortless. Yet, for novices unfamiliar with this type of crafting, there could be some reluctance. Visual representations, rather than mere text, would greatly facilitate their decision-making process."*

Website Functionality Feedback. In evaluating the system's functionality, the BeadMuse AI was described as having an "ease of use." The interviewee noted its customization and flexibility for various proficiency levels as a "significant feature". However, he also highlighted the importance of providing crafting time estimates, elaborating, *"Considering our employees might opt to use this tool during their two-hour lunch breaks, it's essential to account for the captivating nature of iron beads",* and *"it can be challenging to pause once started...having an estimated crafting duration next to each reference could offer valuable guidance."* Beyond the visual cues proposed in the design feedback, he suggested that incorporating a step-by-step tutorial would be beneficial. Such a "beginner tutorial" would aim to introduce beginners to the system's features, ensuring they navigate efficiently.

7.2 Application Scenarios

The interviewee shared several spontaneous scenarios that could potentially incorporate the use of the BeadMuse AI. These first-hand insights offer valuable perspectives on the system's potential integration within the company.

Crafting During Breaks. During the discussion, the interviewee considered the possibility of employees using BeadMuse AI during their daily breaks. He commented, *"Although this might sound presumptuous, I believe many of our employees would want to use BeadMuse AI during their two-hour lunch breaks in the upstairs space."* He also mentioned the enthusiasm displayed by employees who didn't attend the workshop but saw the finished products, *"Many said they wanted to try it in their free time or lunch break, but without templates, they didn't know where to start, but now we can use BeadMuse AI."* He further speculated a considerable number might even engage in crafting at home, seeing it as a "relaxation tool." *"I can see it being something people might spend time on over a few days, just taking it at their own pace."*

Enhancing Employee Creativity Through AI. The interviewee was particularly excited about the AI's ability to produce unexpected images. When he experimented with generating AI images based on text input, the system not only depicted Pikachu in front of iconic American buildings but also cleverly recolored Pikachu with the American flag's hues. He remarked, *"This would be a huge surprise*

for someone like p4. I can imagine they would absolutely love these unexpected images."

8 CONCLUSION AND FUTURE WORK

8.1 Summary of Contributions

We consider our research to contribute to the accessibility in HCI in four aspects: (1)Focus on ID in Accessible Crafting: Prior studies in accessible crafting predominantly centered on physical disabilities. Our research extends this paradigm by specifically addressing individuals with intellectual disabilities, offering a nuanced perspective within this domain. (2)Exploration of Creative Crafting Tool: We investigated iron beads, a distinct creative crafting tool previously rarely examined in the literature. By utilizing this tool for self-expression and initiating workshops, we provided a systematic examination of its applicability and the challenges and opportunities it presents when used by individuals with intellectual disabilities. (3)Development and Preliminary Evaluation of BeadsMuse AI: On the foundation of our empirical insights, we designed the BeadsMuse AI system and undertook a preliminary assessment. Our findings underscore its potential utility in facilitating crafting activities among our target population. (4)Research-Industry Collaboration for Inclusive Employment: By partnering with a company dedicated to inclusive employment, we showcased a way where academia and industry can collaboratively address real-world challenges.

8.2 Limitations and Future Work

Our study, centered on understanding the crafting experiences of individuals with ID, had certain confines. Our participant selection was limited to a singular subsidiary company, potentially limiting the general ability of our findings to a broader context. The study's focus on iron beads, while informed by the known proficiency of the company's employees, means that there's a vast array of crafting activities that remain outside our current purview. The duration of our research provided preliminary insights into the reception and immediate utility of BeadMuse AI. However, its sustained use and long-term benefits necessitate further examination.

Moving ahead, we recognize 3 avenues for exploration and improvement. (1)Enhance BeadMuse AI: Guided by CEO feedback, we plan to enhance the user experience and functional capabilities of BeadMuse AI, making it more intuitive and better tailored to user needs.

(2)Expand the Participant Pool: Our assessments were largely managerial-focused. We aim to extend BeadMuse AI's usage to a broader range of employees, including those with intellectual disabilities, irrespective of their prior workshop attendance.

(3)Long-term Study: In line with our aspiration for BeadMuse AI to serve as a sustained support system for independent crafting among disabled employees, both within our collaborating company and potentially in wider settings, it's imperative to undertake a long-term evaluation. Continuous monitoring over an extended duration will elucidate the system's resilience. Such scrutiny will not only showcase its robust areas but will also identify opportunities for further refinement.

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